Treatment of Gunshot Wounds

To the chest (thoracic cavity):

1. ASSESS THE SCENE! This cannot be stressed enough. If there is still an active shooter, DO NOT APPROACH yet unless you are absolutely sure there is a clear path to remove the PT (e.g. the shooter has left the area).
2. Immediately place a gloved hand over the wound(s) to begin the chest sealing process.
3. If PT needs to be moved, have as many people as possible and practical assist with an emergency move. A backboard (improvised or otherwise) is preferred to help preserve spinal safety.
4. Assess the need for spinal motion restriction.
    a. Manual c-spine should be held regardless, and a c-collar should be applied if available, simply due to the serious mechanism of injury.
5. Completely expose affected area and search for all wounds. You don't want to miss an exit wound. Assess.
    a. If PT presents with difficulty breathing and you hear air movement from a wound, PT is experiencing an open pneumothorax and a chest seal/occlusive dressing will be necessary.
6. Use an occlusive dressing on every wound to the thoracic cavity you see, including exit wounds.
    a. For commercial dressings, try to make sure you have a vented chest seal. If this is the case, the seal can simply be applied directly to the wound.
    b. For a nonvented commercial dressing or an improvised dressing (e.g. plastic bag and duct tape), only secure on three sides. Leave an opening for air to escape the chest.
7. If the PT's difficulty breathing worsens, and they develop hypertension, increasing heart rate, and pale, cool, clammy skin, they are likely developing a tension pneumothorax due to air not escaping properly from the occlusive dressing site.
    a. "burp" the occlusive dressing by peeling it up on one side. You should hear or see pressurized air escape.
    b. If PT is still worsening, a needle decompression may be indicated, HOWEVER IT IS EXTREMELY INADVISABLE THAT YOU PERFORM THIS PROCEDURE. This is typically a paramedic-level skill that should be left to EMS.
8. Ventilating the PT with a BVM if necessary should be performed VERY CAREFULLY. Remember that a pneumothorax PT has decreased lung capacity and ventilating to normal tidal volume will likely cause or worsen a tension pneumothorax if not corrected.
    a. Ventilate 10-12 times a minute with careful attention to chest rise. Cease squeezing the BVM when you observe even minimal chest rise.
9. Monitor PT's SpO2 and other vitals. Monitor for symptoms of hemorrhagic or obstructive shock.

Keep in mind that a bullet will not only cause respiratory symptoms, but obviously will cause internal bleeding and, if it strikes bones, extreme musculoskeletal trauma. In our work environment, the priority is to ensure the PT's breathing isn't compromised so that they'll hopefully survive long enough to make it to EMS. EMS should handle from there, as improper splinting of flail segments can cause severe damage, and improper handling of spinal injury can cause paralysis or death.

See reverse for abdominal and limb gunshot wounds.

To the abdomen:

1. Follow steps 1-5 on reverse side.
2. If organs are eviscerated, DO NOT TOUCH OR REPLACE THE ORGANS.
    a. Follow primary assessment: pay special attention to the PT's airway, as they can suddenly vomit, which will occlude their airway. If PT loses consciousness, an oropharyngeal airway may be indicated.
    b. Position PT with their knees drawn up (sit-up position), as long as their knees and legs are uninjured and spinal injury is not suspected. This will increase pressure on the abdomen and help keep organs contained.
    c. Moisten a large sterile dressing (or dressings) with sterile water or saline. Avoid absorbent gauze or material as this will cling to the organs. Place dressing(s) over wound, covering all organs.
    d. Cover this with an occlusive dressing to retain moisture and warmth. Plastic wrap will do. Secure with tape or cravats.
3. Without evisceration, control bleeding with a gloved hand while preparing dressings.
    a. Monitor for shock throughout treatment
    b. Control external bleeding by applying a dry sterile dressing to wounds.
    c. Position the PT with knees drawn up as long as legs, knees, pelvis, and spine are uninjured.
    d. Do not allow PT to take anything by mouth in any amount, including water, even if the PT complains of thirst. This will help prevent aggravation of organ leakage.

To a limb:

1. Follow steps 1-5 on reverse side.
2. Assess the injured limb. Determine if arterial (bright red, spurting) or venous (dark red, flowing) bleeding is present. If either condition is present, this wound is serious and bleeding out is an urgent concern, especially if the wound is to the femur or other major artery. Death can occur in minutes if bleeding is not managed.
3. Immediately apply direct pressure with a gloved hand while a dressing is prepared.
4. Apply a dry sterile dressing to the wound and secure, continuing direct pressure.
5. If bleeding doesn't stop, a tourniquet is indicated.
    a. Apply the tourniquet proximal to the wound (between the wound and the torso), but as close as possible. DO NOT APPLY A TOURNIQUET OVERTOP A JOINT.
    b. Tighten the tourniquet in the manufacturer-prescribed manner (windlass, loops, etc.) until no distal pulse (pedal for a leg wound, radial for an arm wound) can be felt. The active bleeding should stop at this point. If it doesn't, a second tourniquet may be indicated. Apply this one proximal to the first tourniquet.
    c. If possible, record the time that the tourniquet was applied and pass this info to EMS. Tourniquets can only be applied for so long before the cause more damage than they fix. However, keep "life over limb" in mind. Don't hesitate to apply a tourniquet to an uncontrolled serious bleed because you fear damage from the tourniquet.

See reverse for chest gunshot wounds.